

KOMMUNAR, Russian Federation

WMO: 297590

Lat: 54.333N

Lon: 89.283E

Elev: 847

StdP: 91.56

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.4	-27.7	-33.3	0.2	-30.3	-30.7	0.2	-27.6	7.5	-14.0	6.5	-13.2	0.7	270	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	25.8	16.8	23.9	16.1	22.1	15.2	18.1	23.6	17.1	22.3	16.1	20.8	1.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.7	20.3	15.1	11.9	19.2	14.2	11.2	18.5	54.8	23.4	51.4	22.3	48.5	20.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.5	3.8	-32.1	29.5	4.2	1.9	-35.1	30.9	-37.6	32.0	-39.9	33.1	-42.9	34.5		
(5)			-32.3	20.3	4.2	1.5	-35.3	21.4	-37.8	22.3	-40.1	23.1	-43.1	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-15.1	-12.7	-5.9	1.2	7.0	14.0	15.4	13.4	7.2	0.1	-8.1	-12.9	(6)
(7)		DBStd	11.98	6.65	6.44	5.93	5.61	5.11	4.07	3.03	3.71	4.57	5.31	6.68	6.24	(7)
(8)		HDD10.0	3991	779	636	493	266	122	13	1	11	109	309	544	709	(8)
(9)		HDD18.3	6576	1037	869	752	514	351	136	98	157	336	566	794	967	(9)
(10)		CDD10.0	477	0	0	0	2	30	134	169	117	23	2	0	0	(10)
(11)		CDD18.3	21	0	0	0	0	1	6	8	6	1	0	0	0	(11)
(12)		CDH23.3	221	0	0	0	0	8	82	76	49	7	0	0	0	(12)
(13)		CDH26.7	29	0	0	0	0	1	9	10	8	1	0	0	0	(13)
(14)	Wind	WSAvg	1.2	1.4	1.4	1.4	1.2	1.1	0.9	0.7	0.7	0.9	1.2	1.5	1.8	(14)
(15)	Precipitation	PrecAvg	822	41	36	55	68	68	77	85	82	66	82	76	67	(15)
(16)		PrecMax	1050	131	95	149	144	152	146	212	158	116	176	153	160	(16)
(17)		PrecMin	605	0	0	14	16	14	25	42	20	15	21	42	11	(17)
(18)		PrecStd	119	30	24	35	33	37	27	37	27	26	35	29	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	2.4	12.2	18.3	24.3	28.0	28.1	27.6	24.0	16.7	8.0	0.1	(19)
(20)			MCWB	-1.2	-0.9	4.7	8.5	13.3	17.0	18.3	17.9	14.1	8.3	3.1	-1.7	(20)
(21)		2%	DB	-2.2	-0.6	7.5	14.9	21.3	25.6	25.5	24.3	19.9	12.9	4.3	-1.8	(21)
(22)			MCWB	-3.6	-2.9	2.1	7.0	12.1	16.1	17.5	16.8	12.2	7.3	2.0	-3.1	(22)
(23)		5%	DB	-4.1	-2.4	4.6	12.2	18.7	23.6	23.5	22.0	16.6	10.0	1.8	-3.4	(23)
(24)			MCWB	-5.5	-4.2	0.5	5.5	10.2	15.6	16.7	15.7	10.8	5.8	0.1	-4.7	(24)
(25)		10%	DB	-6.3	-3.9	2.2	9.7	16.0	21.7	21.6	19.9	14.2	7.6	0.0	-4.8	(25)
(26)			MCWB	-7.5	-5.5	-0.6	4.4	9.2	14.7	15.9	14.8	10.0	4.6	-1.2	-6.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-0.2	5.1	9.3	14.6	18.7	19.7	19.1	15.2	9.4	4.9	-1.0	(27)
(28)			MCDWB	0.2	1.6	11.5	16.7	22.6	25.2	25.3	25.2	21.4	14.3	6.7	-0.3	(28)
(29)		2%	WB	-3.3	-2.6	2.5	7.6	12.6	17.4	18.3	17.4	13.0	7.7	2.0	-2.9	(29)
(30)			MCDWB	-2.5	-0.9	6.5	13.4	19.4	23.3	23.6	22.8	18.1	11.8	4.2	-1.9	(30)
(31)		5%	WB	-5.5	-4.1	0.9	6.2	11.2	16.4	17.4	16.3	11.6	6.2	0.3	-4.6	(31)
(32)			MCDWB	-4.1	-2.5	4.1	11.0	16.7	22.1	22.3	20.9	15.6	9.7	1.7	-3.6	(32)
(33)		10%	WB	-7.4	-5.5	-0.6	4.8	9.9	15.3	16.4	15.3	10.3	4.6	-1.3	-6.0	(33)
(34)			MCDWB	-6.3	-4.0	2.2	9.1	14.9	20.3	20.6	19.3	13.6	7.4	0.0	-4.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	8.0	9.5	9.7	9.9	10.5	9.1	9.2	8.8	6.9	5.9	6.0	(35)
(36)			MCDBR	7.2	8.5	12.9	13.7	14.9	14.5	12.8	13.1	13.8	11.2	6.8	6.7	(36)
(37)			MCWBR	6.2	6.7	8.6	7.7	7.4	6.8	6.0	6.3	7.2	6.8	5.1	5.9	(37)
(38)		5% WB	MCDBR	7.2	7.8	12.1	12.7	12.4	12.2	10.9	11.5	11.9	10.4	6.5	6.6	(38)
(39)			MCWBR	6.3	6.3	8.5	7.5	6.7	6.1	5.7	6.0	6.7	6.7	5.1	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.252	0.285	0.331	0.372	0.379	0.376	0.390	0.376	0.342	0.321	0.284	0.252	(40)	
(41)		taud	2.115	2.087	2.024	2.134	2.257	2.362	2.333	2.349	2.397	2.311	2.118	2.044	(41)	
(42)		Ebn at Noon	691	790	827	848	865	871	849	840	826	748	628	586	(42)	
(43)		Edh at Noon	65	96	129	134	126	115	116	108	90	77	65	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.91	3.26	4.45	5.08	5.72	5.10	4.27	2.94	1.59	0.83	0.59	(44)	
(45)		RadStd	0.12	0.19	0.28	0.26	0.63	0.55	0.35	0.41	0.34	0.17	0.13	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page